Math Training — M1 Cognitive Science

Session 3 — Analysis I

Functions, derivatives, integrals, differential equations

Advanced track

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What today is about

Today is about **functions of one variable**: what they are, how to measure the way they change, and how to add change up again.

- **Functions**: the handful of shapes that keep coming back.
- **Derivatives**: the slope, the rules, and what the *sign* tells you.
- **Integrals**: area under a curve, and the antiderivative.
- **A first differential equation**: what it means to *solve* one, and why the exponential is the answer.

Next session does the same for functions of *several* variables.

The one idea to hold onto

If you remember only one thing

A **derivative** says how fast something is changing right now. An **integral** adds those changes back up. A **differential equation** tells you the rate in terms of the thing itself — and to *solve* it is to produce the function that does what the equation says.

That last sentence is also a free check on every answer you will ever write: substitute it back and see whether both sides agree.

What a function is

Definition

A **function** f is a rule that turns one number into another: give it t, it returns $f(t)$. Its **graph** is what you get by marking the height $f(t)$ above each t.

Two words worth keeping straight:

- the **variable** is what you feed in — time, task difficulty, number of practice trials;
- the **parameters** are the numbers that fix *which* function you have — and they do not change as the variable runs.

In $f(t) = a\,\mathrm{e}^{-t/\tau}$, the variable is t; a and τ are parameters.

Five shapes cover almost everything

- **Constant** $f(t) = c$ — a flat line. Nothing changes.
- **Linear** $f(t) = mt + c$ — a straight line, climbing m per unit, starting at c.
- **Power** $f(t) = t^n$ — and n may be negative or fractional: $1/t = t^{-1}$, $\sqrt{t} = t^{1/2}$.
- **Exponential** $f(t) = \mathrm{e}^{at}$ — growth if $a > 0$, decay if $a < 0$. The shape this whole course runs on.
- **Logarithm** $f(t) = \ln t$ — the inverse of the exponential: it undoes it.

The sine appears too, but only in Session 4.

Definition

$\mathrm{e} \approx 2.718$ is the base for which the curve e^t climbs, at every point, at exactly the height it has reached. For $a < 0$ write $a = -1/\tau$:

$$f(t) = \mathrm{e}^{-t/\tau},$$

and the positive number τ is the **time constant**.

Key idea

After one τ, the quantity has fallen to $\mathrm{e}^{-1} \approx 37\%$ of where it started — and after another τ to 37% of *that*, for ever.

Three numbers worth memorising: $\mathrm{e}^{-1} \approx 0.37$, $\mathrm{e}^{-2} \approx 0.14$, $\mathrm{e}^{-3} \approx 0.05$. So “about three time constants” is the practical answer to “when is it over”.

The forgetting curve

- ### Link to cognitive science

  Learn a list of words, then test recall after a delay t. The proportion still recalled falls off as

  $$R(t) = R_0\,\mathrm{e}^{-t/\tau},$$

  the shape Ebbinghaus measured on himself in 1885 and the standard first description of forgetting ever since. Here R_0 is what you recall immediately and τ is how long the memory "lasts" in the sense above.

With $R_0 = 0.9$ and $\tau = 2$ days: after 2 days $0.9 \times 0.37 \approx 0.33$; after 6 days $0.9 \times 0.05 \approx 0.05$. Two parameters, and the whole curve.

Real forgetting data are often fitted better by a power law $R(t) = R_0\,t^{-b}$ — worth one sentence, because it is a live debate and because it is a different one of our five shapes.

Practice — try it

1. Sketch $f(t) = \mathrm{e}^{-t/4}$ and $g(t) = \mathrm{e}^{-t}$ on the same axes. Which falls faster, and why?
2. Recall follows $R(t) = 0.8\,\mathrm{e}^{-t/3}$ with t in days. Give $R(0)$, $R(3)$, $R(9)$.
3. Simplify $\ln(\mathrm{e}^5)$, $\mathrm{e}^{\ln 7}$, $\ln(a^2 b)$.
4. After how long has R fallen to half its initial value? (Solve $\mathrm{e}^{-t/\tau} = \frac{1}{2}$.)

The derivative is a slope

Definition

For a straight line, "how fast is it climbing" is the **slope**: rise over run. For a curve the slope changes from point to point, so we do the same thing *locally* — zoom in far enough and any smooth curve looks straight. The slope of that line is the **derivative**, written $f'(t)$ or $\dfrac{df}{dt}$:

$$f'(t) = \lim_{h \to 0} \frac{f(t+h) - f(t)}{h}.$$

When the variable is time, the derivative is a *rate of change per unit time*. When it is anything else — difficulty, dose, number of trials — it is still "how much does the output move when I nudge the input".

Useful fact

$$c \to 0, \qquad t^{\,n} \to n\,t^{\,n-1}, \qquad \mathrm{e}^{at} \to a\,\mathrm{e}^{at}, \qquad \ln t \to \frac{1}{t}$$

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \qquad (fg)' = f'g + fg', \qquad \big(f(g(t))\big)' = f'(g(t)) \cdot g'(t)$$

The power rule holds for negative and fractional n too: $1/t \to -1/t^2$ and $\sqrt{t} \to 1/(2\sqrt{t})$.

The last one is the **chain rule**: differentiate the outside, then multiply by the derivative of the inside. It is where $\mathrm{e}^{at} \to a\,\mathrm{e}^{at}$ actually comes from.

The mistake to head off

Watch out

$\mathrm{e}^{-t/2}$ is *not* a power of t, so the power rule has nothing to say about it. Its derivative is $-\frac{1}{2}\,\mathrm{e}^{-t/2}$.

Answers shaped like $(-t/2)\,\mathrm{e}^{(-t/2)-1}$ come from applying $t^{\,n} \to n\,t^{\,n-1}$ to an exponential — treating the exponent as if it were the power.

The difference matters: in $t^{\,n}$ the variable is in the *base*, in e^{at} it is in the *exponent*. Ask which rule applies before anyone writes anything down, and do it every time an exponential appears today.

What the sign tells you

$$f' > 0 \;\Rightarrow\; \text{rising}, \qquad f' < 0 \;\Rightarrow\; \text{falling}, \qquad f' = 0 \;\Rightarrow\; \text{momentarily flat}$$

— a peak, a trough, or a value that has stopped moving.

- **Intuition**

  To find where a function is largest: set $f' = 0$, solve, and check the sign of f' on either side. Positive then negative means a maximum; negative then positive, a minimum.

Very often the sign is all you need, and you never have to evaluate anything.

Example

Differentiate $f(t) = 3t^2 - 4t + 7$, $R(t) = 0.9\,\mathrm{e}^{-t/2}$, $\mathrm{RT}(n) = 800\,n^{-0.3}$.

Polynomial, term by term: $3t^2 \to 6t$, $-4t \to -4$, $7 \to 0$, so $f'(t) = 6t - 4$. It vanishes at $t = \frac{2}{3}$, negative before and positive after: a minimum.

Forgetting curve: $a = -\frac{1}{2}$, so $R'(t) = -\frac{1}{2} \cdot 0.9\,\mathrm{e}^{-t/2} = -\frac{1}{2}R(t)$. Note what this says: *the rate of forgetting is proportional to how much you still remember* — that sentence is a differential equation, and we come back to it.

Power law of practice: $\mathrm{RT}'(n) = 800 \times (-0.3)\,n^{-1.3} < 0$ — reaction time keeps falling with practice, but the improvement per extra trial shrinks fast.

Practice — try it

1. Differentiate $f(t) = t^2 - 4t + 7$, $g(t) = 5\,\mathrm{e}^{-2t}$, $h(t) = 1/t + \sqrt{t}$.
2. Differentiate $R(t) = R_0\,\mathrm{e}^{-t/\tau}$ and show that $R' = -R/\tau$.
3. Use the chain rule on $f(t) = \mathrm{e}^{-t^2}$ and on $g(t) = \ln(1 + 3t)$.
4. Use the product rule on $f(t) = t\,\mathrm{e}^{-t}$. Where is its maximum?

The integral adds the changes back up

Definition

$\int_a^b f(t)\,dt$ is the **signed area** between the graph of f and the horizontal axis, from $t = a$ to $t = b$ — area above the axis counting positive, below it negative.

Two readings, both useful:

- chop $[a, b]$ into thin slices of width dt, each contributing $f(t)\,dt$, and add them up. That is what the elongated S means: a sum.
- if f is a *rate*, the integral is the *total accumulated*. Speed integrates to distance; a rate of responding integrates to a number of responses.

Useful fact

If $F' = f$, then

$$\int_a^b f(t)\,dt = F(b) - F(a).$$

F is called an **antiderivative** of f, and it is unique only up to an additive constant — hence the "$+C$" in the indefinite integral $\int f(t)\,dt = F(t) + C$.

So integration is differentiation run backwards, and the table of derivatives becomes a table of integrals read right to left:

$$\int t^n dt = \frac{t^{n+1}}{n+1}\ (n \neq -1), \qquad \int \frac{dt}{t} = \ln|t|, \qquad \int \mathrm{e}^{at} dt = \frac{\mathrm{e}^{at}}{a}.$$

Two techniques, and that is all we need

- **Substitution — the chain rule backwards**

$$\int f(g(t))\, g'(t)\, dt = \int f(u)\, du \quad \text{with} \quad u = g(t), \;\; du = g'(t)\, dt.$$

- **Integration by parts — the product rule backwards**

$$\int u\, v'\, dt = uv - \int u' v\, dt.$$

Choosing u in the second: pick the factor that gets *simpler* when differentiated. A polynomial times an exponential is the standard case — the polynomial is u, and each application drops its degree by one.

Example

Compute $\displaystyle\int_0^T e^{-t/\tau} dt,\ \int 2t\, e^{t^2} dt,\ \int_0^\infty t\, e^{-t} dt.$

Directly: $\int_0^T e^{-t/\tau} dt = \left[-\tau\, e^{-t/\tau}\right]_0^T = \tau\left(1 - e^{-T/\tau}\right)$. As $T \to \infty$ this tends to τ: the total area under a decaying exponential is exactly its time constant.

By substitution: $u = t^2$, $du = 2t\, dt$, so the integral is $\int e^u du = e^{t^2} + C$.

By parts: $u = t$, $v' = e^{-t}$, so $\int_0^\infty t\, e^{-t} dt = \left[-t\, e^{-t}\right]_0^\infty + \int_0^\infty e^{-t} dt = 0 + 1 = 1.$

Practice — integrals

Practice — try it

1. Compute $\int_0^2 (3t^2 - 4t + 1)\,dt$ and $\int_1^{\mathrm{e}} \frac{dt}{t}$.
2. Compute $\int_0^\infty \mathrm{e}^{-3t} dt$. Interpret the answer.
3. By substitution: $\int \frac{2t}{1+t^2}\,dt$.
4. By parts: $\int t\,\mathrm{e}^{-2t} dt$.

Solving is not integrating

Definition

If $\dot{x} = f(t)$, the rate depends on *time*, and you integrate: $x(t) = \int_0^t f(s)\,ds + x_0$. Nothing is circular — the right-hand side never mentions x.

If $\dot{x} = f(x)$, the rate depends on *the unknown itself*. You cannot integrate the right-hand side, because you do not yet know $x(t)$ to integrate.

Only the second is a differential equation proper, and it needs a different move.

The error to name out loud

Watch out

This one is made by the *strong* students. Asked to solve $\dot{x} = -(x-3)$, they anti-differentiate the right-hand side and return $-x^2/2 + 3x$. That is an antiderivative of something; it is not a solution of anything.

Two sentences fix it for good.

First: an antiderivative of a function of x is another function of x, but the answer you want is a function of t — the types do not even match.

Second: substitute and check. By the chain rule $\frac{d}{dt}\big(-x^2/2 + 3x\big) = (-x+3)\,\dot{x}$, which equals $-(x-3)$ only if $\dot{x} = -1$ — not what was asked.

The equation the whole course rests on

- **Definition**

$$\frac{dx}{dt} = ax + b \qquad \Longrightarrow \qquad x(t) = x^\star + (x_0 - x^\star)\,\mathrm{e}^{at}, \qquad x^\star = -\frac{b}{a} \;\; (a \neq 0),$$

where $x^\star$ is the **steady state** (the rate vanishes there) and $\tau = -1/a$ is the **time constant**.

Read it as a sentence rather than a formula: *start at x_0, end at $x^\star$, and close the gap between them exponentially.* One τ closes 63% of the gap; three close about 95%.

If $a = 0$ the equation is just $\dot{x} = b$, there is no steady state, and $x(t) = x_0 + bt$ marches off linearly for ever. That is why $a \neq 0$ is written.

Stability is the sign of a slope

- **Intuition**

  The rate, plotted against x, is the straight line $ax + b$, and its slope is a. Perturb the steady state by δ: then $\dot{\delta} = a\delta$, which decays if and only if $a < 0$.

$$a < 0 \;\Rightarrow\; \textbf{stable}\text{: the system returns,} \qquad a > 0 \;\Rightarrow\; \textbf{unstable}\text{: it runs away.}$$

There is a picture that carries the same information without any algebra: mark $x^\star$ on a line and put an arrow at each side pointing the way the rate says. Arrows pointing in, stable. Arrows pointing out, unstable. It is called the **phase line**, and unlike the formula it still works when the rate is a curve.

Link to cognitive science

Performance P improves with practice, but not without limit: there is a ceiling $P_{\max}$, and the closer you get the slower you improve. Written as an equation,

$$\frac{dP}{dt} = k\,(P_{\max} - P),$$

which is $\dot{x} = ax + b$ with $a = -k$ and $b = kP_{\max}$, so $P^\star = P_{\max}$ and

$$P(t) = P_{\max} - (P_{\max} - P_0)\,\mathrm{e}^{-kt}.$$

With $P_0 = 20$, $P_{\max} = 100$, $k = 0.1$ per session: after 10 sessions $P = 100 - 80 \times 0.37 \approx 70$; after 30, $P \approx 96$. The time constant is $\tau = 1/k = 10$ sessions.

Note that this is the forgetting curve upside down — same equation, same exponential, opposite direction. That is not a coincidence: both say "the rate is proportional to the distance still to go".

Simulating: Euler's method

Definition

$$x(t + \Delta t) \;\approx\; x(t) + \Delta t \cdot \dot{x}(t).$$

In words: look up how fast you are moving, move at that speed for a short while, then look again.

The step pretends the rate is constant across the interval, so the error over a fixed duration is proportional to Δt — halve the step and you roughly halve the error.

Watch out

The step cannot be arbitrarily large: for $\dot{x} = ax$ with $a < 0$ the scheme *blows up* unless $\Delta t < 2/|a| = 2\tau$. And stability is not accuracy — staying below 2τ only stops the simulation exploding.

Practice — a first differential equation

Practice — try it

1. Which are integration problems and which are differential equations? (i) $\dot{x} = 3t^2$; (ii) $\dot{x} = -2x$; (iii) $\dot{x} = -2x + 5$. Solve all three with $x_0 = 1$.
2. For $\dot{x} = -x + 4$: give $x^\star$ and τ, decide stability, and describe the trajectory from $x_0 = 10$ and from $x_0 = 0$.
3. Recall obeys $\dot{R} = -R/3$ with $R_0 = 0.8$ (t in days). Write $R(t)$ and give $R(6)$.
4. Take one Euler step of $\dot{x} = -x + 2$ from $x = 0$ with $\Delta t = 0.1$. What is the largest step that would not blow up?

Session 3 — what we built

- A **function** is a rule with a variable and some parameters; five shapes cover almost everything, and $\mathrm{e}^{-t/\tau}$ is the one that matters most.
- The **derivative** is a slope. Sign: rising, falling, or flat. Rules: power, exponential, sum, product, chain.
- The **integral** is a signed area, and the fundamental theorem makes it differentiation run backwards. Substitution and by parts are the two techniques we need.
- $\dot{x} = ax + b$ starts at x_0, ends at $x^\star = -b/a$, and closes the gap like e^{at}. *Solving* it is not *integrating* it, and stability is the sign of a.

Next session: functions of *several* variables — surfaces, partial derivatives, gradients — and functions whose output is a *vector*.